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Dislocation-grain boundary interaction in metallic materials: Competition between dislocation transmission and dislocation source activation

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ABSTRACT

Dislocation transmission across grain boundary (GB) and activation of dislocation source in adjacent grains are considered common features of dislocation-GB interactions in regulating mechanical properties and behaviors of metallic materials and structures. However, it is currently unclear which of dislocation transmission and dislocation source activation plays the dominant role in regulating dislocation-GB interaction. To study the competition between dislocation transmission and dislocation source activation, a theoretical framework is established based on a dislocation pile-up model. It is found that both dislocation transmission and dislocation source activation exhibit strong dependence on the GB misorientation angle. The critical transmission stress derived based on an energy method also depends on the grain size in a scaling form similar to the Hall-Petch relation. The outgoing slip systems during dislocation transmission are successfully predicted and agree with experimental observations. Regarding the dislocation source activation, the critical activation stress is investigated by examining the local stress fields generated by single and multiple slip dislocation pile-ups. It is found that compared with single slip dislocation pile-up, the result of multiple slip dislocation pile-ups displays lower critical activation stresses and exhibits feature of sensitivity to loading direction, interpreting the reported results of dislocation source activation in polycrystals. Further comparison between dislocation transmission and dislocation source activation for ⟨100⟩ tilt GBs indicates that the dislocation transmission prevails at low angle GB, while dislocation source activation is prone to occur at high angle GB. Our theoretical studies quantify the dislocation-GB interaction induced by dislocation pile-ups, shed light on the competition between dislocation transmission and dislocation source activation, and might provide essential guidance for high-performance metallic material design.

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1. Introduction

Improving mechanical properties of crystalline metallic materials has been one of the eternal pursuits in materials science. In recent years, plainification has been proposed to provide crystalline metallic materials with high strength and excellent ductility without changing the chemical composition (Li and Lu, 2019). For example, nanocrystalline metals with bimodal microstructures (Wang et al., 2002), nanotwinned materials (Lu et al., 2009) and gradient structure materials (Lin et al., 2018) are three successful cases, in which excellent mechanical properties are achieved by controlling microstructures and spatial distribution of grain boundaries (GBs). From a microscopic perspective, the interaction between dislocations and GBs, including dislocation transmission across GBs (Bieler et al., 2019; Guo et al., 2015; Lee et al., 1990) and activation of dislocation sources in adjacent grains (Bayerschen et al., 2016; Cottrell, 1958; Genée et al., 2017), shall play an important role in affecting the mechanical behaviors and properties of crystalline metallic materials, which could further assist crack initiation (Li et al., 2013; Shanthraj and Zikry, 2013; Zhang and Wang, 2003), crack propagation (Chen et al., 2017; Wilson et al., 2018), and cavity nucleation (Lim, 1989; Yoo and Trinkaus, 1986).

It is well known that the dislocation-GB interaction is regulated by the stress concentration arising from dislocation pile-ups, whose formation is caused by GBs hindering the motion of dislocations (Ding et al., 2016; Gemperlová et al., 2002; Hirth and Lothe, 1982; Hull et al., 2011), and has been theoretically analyzed. For example, Eshelby et al. (1951) established a discrete dislocation model to analyze the dislocation distribution in a pile-up. However, a lack of analytical solutions from that discrete model limits its applications in scientific and engineering studies. Later on, a continuum model has been proposed (Hirth and Lothe, 1982; Leibfried, 1951), from which analytical solutions of the dislocation distribution and stress concentration in the dislocation pile-up have been derived. These analytical solutions are further verified by the dislocation dynamics (DD) simulations (Quek et al., 2014). The dislocation-GB interaction is also closely related to the misorientation between two adjacent grains (Shen et al., 1986; 1988). As an important mechanism of dislocation-GB interaction, the dislocation transmission has been a continuous focus of interest for mechanical engineers and material scientists. Nowadays, there are two mostly used empirical methods to qualitatively assess the role of microstructures in dislocation transmission. One is the plastic compatibility factor developed by Luster and Morris (1995) taking into account the spatial information of slip systems. The larger the plastic compatibility factor is, the easier the dislocation transmission could occur. The other method is based on the LRB criteria established by Lee et al. (1989), that is, (1) the misorientation angle between the slip planes should be minimized; (2) the resolved shear stress (RSS) on the outgoing slip system arising from the dislocation pile-up should be maximized; (3) the magnitude of the Burgers vector of residual dislocation should be minimized. Although both methods provide convenient and effective assessment of the likelihood of dislocation transmission (Bayerschen et al., 2016; Han et al., 2018; Hémery et al., 2018; Kacher et al., 2014), the related micro-mechanism was veiled because only qualitative analysis can be performed. Currently, further explorations with molecular dynamics (MD) studies (Adlakha and Solanki, 2016; Sangid et al., 2011; Tsuru et al., 2009) and DD simulations (Kapoor and Verdhan, 2017; Li et al., 2009; Liu et al., 2012; Lu et al., 2019) provided physical insight into the dislocation transmission and revealed the role of GB interfacial energy in regulating the dislocation transmission. Though the dislocation transmission has been extensively studied, it remains challenging to quantitatively assess the dislocation transmission at various GB misorientation angles and clarify its role in affecting the yield strength of crystalline metals.

In addition to dislocation transmission across GBs, dislocation pile-ups could lead to the activation of dislocation sources in adjacent grains to assist plastic deformation of polycrystals (Cottrell, 1958; Davis et al., 1966; Hémery et al., 2018; Joseph et al., 2018). Experiments have shown that the adjacent grains undergo serious plastic deformation even in the absence of dislocation transmission (Genée et al., 2017; Joseph et al., 2018), indicating that the dislocation sources are certainly activated in adjacent grains. Due to the dislocation pile-up, the stress by superposition of Peach–Koehler forces can activate the dislocation source in adjacent grains (Guo et al., 2014; Hirth and Lothe, 1982; Quek et al., 2014). The model proposed by Livingston and Chalmers (1957) has been applied to calculate the projected stress on the outgoing slip system arising from the dislocation pile-up in one slip system. However, that model is insufficient to describe the behavior of dislocation source activation (Davis et al., 1966), and the quantitative relation between critical activation stress and GB misorientation angles is not established yet.

In this study, a theoretical framework taking into account both dislocation transmission and dislocation source activation is established to investigate the underlying physics of the dislocation-GB interaction. The critical stress of dislocation transmission is formulated based on the energy release rate during the dislocation pile-up formation and the energy barrier for dislocation transmission. Moreover, dislocation source activation for multiple slip systems is characterized by incorporating the Frank-Read (FR) source model (Foreman, 1967). It is found that the activation of dislocation sources is sensitive to the loading direction, consistent with recent experimental results (Genée et al., 2017). Finally, the comparison of critical stresses between the dislocation transmission and the dislocation source activation is performed at low angle GB (LAGB) and high angle GB (HAGB). Our results help to improve the understanding of the dislocation pile-up based dislocation-GB interaction and provide the corresponding quantitative descriptions to assist rational design of metallic materials with excellent mechanical properties.

The remainder of this paper is organized as follows. In Section 2, the dislocation pile-up is modelled and types of dislocation-GB interaction are introduced. In Section 3, the residual dislocation tensor is introduced and the critical stress of dislocation transmission is determined from an energy-based method. In Section 4, the activation of dislocation sources in adjacent grains is studied under single and multiple slip dislocation pile-ups. Theoretical results are presented to de-

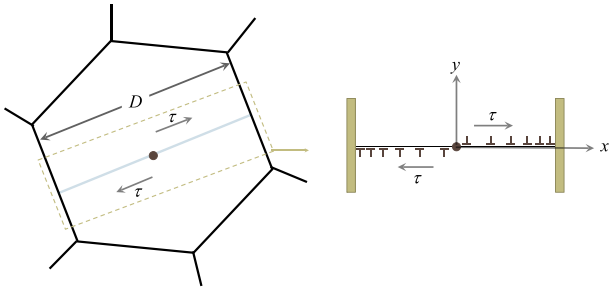

Fig. 1. Schematic of dislocation pile-ups at GBs under applied stress.

scribe the dislocation transmission and dislocation source activation, respectively, in Sections 5 and 6, which are followed by conclusions in Section 7.

2. Formation of dislocation pile-up and associated stress concentration

2.1. Modeling dislocation pile-up formation

The dislocation motion is strongly affected by GBs, leading to the stress concentration associated with the formation of dislocation pile-ups. In this study, a dislocation pile-up model with the concept of continuous dislocation distribution is applied to analyze the dislocation distribution and associated stress fields (Hirth and Lothe, 1982; Leibfried, 1951). In our continuum model with the adopted Cartesian coordinate in Fig. 1, the equilibrium position of dislocations is described by the equilibrium linear dislocation density $n(x)$, which represents the number of dislocations per unit length. The dislocations in the slip system are subjected to the external applied stress and the Peach-Koehler force from other dislocations, and the following equilibrium equation is required

$$\tau_{\text{eff}} b + \frac{\mu b^2}{2\pi(1-\nu)} \int_{-D/2}^{D/2} \frac{n(x')}{x-x'} \mathrm{d}x' = 0, \qquad x \in \left(-\frac{D}{2}, \frac{D}{2}\right) \tag{2.1}$$

with $\tau_{\text{eff}} = \max(\tau - \tau_0, 0)$ as the effective stress associated with the dislocation motion, τ as the RSS, and τ_0 as the resistance stress arising from the lattice friction, dislocation forest hardening and so on. In Eq. (2.1), μ is the shear modulus, ν is the Poisson ratio, b is the magnitude of Burgers vector, D is the grain size, and the linear dislocation density is defined as $n(x) = (\mathrm{d}b/\mathrm{d}x)/b$.

The analytical solution of the singular integral Eq. (2.1) can be obtained using Hilbert transforms and represented in terms of Tschebyscheff polynomials (Akarapu and Hirth, 2013; Hirth and Lothe, 1982; Leibfried, 1951; Liu et al., 2014). Implementing the condition that the dislocation source always produces the same number of positive and negative dislocations

$$N_{\text{total}} = \int_{-D/2}^{D/2} n(x)\mathrm{d}x = 0, \tag{2.2}$$

the solution of Eq. (2.1) is obtained as

$$n(x) = \frac{2(1-\nu)\tau_{\text{eff}}}{\mu b} \frac{x}{\sqrt{(D/2)^2 - x^2}}, \qquad x \in \left(-\frac{D}{2}, \frac{D}{2}\right). \tag{2.3}$$

According to the Taylor-Orowan relation, the total shear strain is

$$\varepsilon = \int_{-D/2}^{D/2} \frac{bxn(x)}{D} \mathrm{d}x = \frac{\pi(1-\nu)D\tau_{\text{eff}}}{4\mu}. \tag{2.4}$$

The total potential energy Π per unit length (along the dislocation line) of the dislocation pile-up is a sum of the elastic energy W due to the dislocation interaction and the potential energy V due to the external force, and can be expressed as

$$\Pi = W + V = D\int_0^{\varepsilon} \tau'_{\text{eff}} \mathrm{d}\varepsilon' - D\tau_{\text{eff}}\varepsilon = -\frac{\pi(1-\nu)D^2 {\tau_{\text{eff}}}^2}{8\mu}. \tag{2.5}$$

Consequently, the energy release rate in one end of the dislocation pile-up during the leading dislocation motion (i.e. dislocation transmission) is

$$G = -\frac{\partial \Pi}{\partial D} = \frac{\pi(1-\nu)D{\tau_{\text{eff}}}^2}{4\mu}, \tag{2.6}$$

which is suitable for energy-based analysis of dislocation transmission. Eq. (2.6) has a form similar to that of the energy release rate at a shear crack tip (Anderson, 2005).

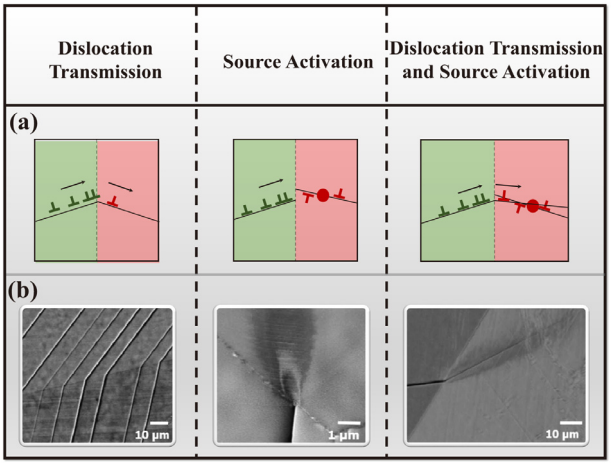

Fig. 2. Dislocation-GB interaction. (a) Schematics of the interaction mechanisms and (b) experimental demonstration (Genée et al., 2017).

In addition, the shear stress $\tau_{xy}(x, 0)$ in adjacent grains generated by dislocation pile-ups can be determined by the superposition of shear stresses produced by each dislocation (Hirth and Lothe, 1982)

$$\tau_{xy}(x,0) = \int_{-D/2}^{D/2} \frac{\mu b}{2\pi(1-\nu)} \frac{n(x')}{x-x'} \mathrm{d}x' = \tau_{\mathrm{eff}} \left[\frac{x}{\sqrt{x^2-(D/2)^2}} - 1 \right]. \tag{2.7}$$

In the vicinity of the GB ($x - D/2 \ll D/2$), the shear stress is estimated as

$$\tau_{xy}(x,0) \cong \frac{\tau_{\mathrm{eff}}}{2} \sqrt{\frac{D}{x-D/2}}. \tag{2.8}$$

The stress concentration near the GB ($x \approx D/2$) has a similar form to the stress field at the crack tip, which has been identified by experiments and simulations (Britton and Wilkinson, 2012; Genée et al., 2017; Guo et al., 2014; Quek et al., 2014). Both the energy release rate in Eq. (2.6) and stress field given by Eq. (2.7) regulate the dislocation-GB interaction.

2.2. Dislocation-GB interaction

Experimental observations suggest that the underlying mechanism of dislocation-GB interaction consists of dislocation transmission and dislocation source activation (Bayerschen et al., 2016), and three situations might exist as shown in Fig. 2. As the dislocation transmits across the GB, the dislocation pile-up could be relaxed (Haouala et al., 2020), which means that the stress concentration near the GB would not be further strengthened. Therefore, the occurrence of dislocation transmission is unfavorable to the activation of dislocation source in adjacent grains. This deduction is consistent with statistical experimental results that dislocation transmission mostly occurs at LAGB, and dislocation source activation is more likely to occur at HAGB (Bayerschen et al., 2016; Genée et al., 2017; Hémery et al., 2018). In the following sections, we introduce how to determine the critical stress required for the occurrence of these situations based on the dislocation pile-up model.

3. Dislocation transmission and residual dislocations

3.1. A criterion of dislocation transmission across GBs

In previous studies (Han et al., 2018; Li et al.,2009; Ma et al., 2006; Sangid et al., 2011), the dislocation transmission has been investigated based on an energy method requiring a negative change of the total potential energy $\Delta\Pi$, that is,

$$\Delta\Pi = E_{\mathrm{barrier}} - E_{\mathrm{release}} \le 0, \tag{3.1}$$

where E_{barrier} is the energy barrier for dislocation transmission and E_{release} is the corresponding released energy of the dislocation pile-up.

According to Eq. (2.6), E_{release} can be estimated as

$$E_{\mathrm{release}} = \int_{D}^{D+b/p} G\mathrm{d}D' \approx \frac{Gb}{p} = \frac{\pi(1-\nu)D\tau_{\mathrm{eff}}{}^2 b}{4\mu p}, \tag{3.2}$$

where G is the energy release rate in one end of the dislocation pile-up and b/p is assumed to be the required displacement of the leading dislocation to transmit across the GB, which means that the displacement component perpendicular to the

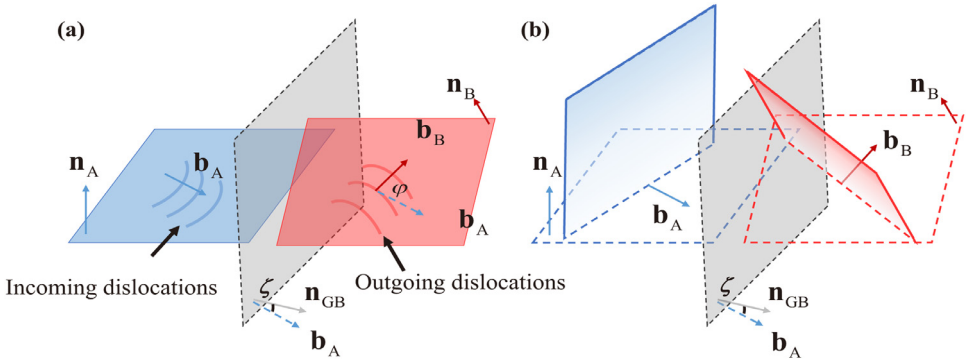

Fig. 3. Schematic diagram of edge dislocation transmission across the grain boundary. (a) Incoming and outgoing slip systems, (b) extra half-plane of atoms for edge dislocations on incoming and outgoing slip systems.

GB has a magnitude of Burgers vector. Here p is the dislocation-GB interaction coefficient defined by

$$p = \frac{|\mathbf{b}_A \cdot \mathbf{n}_{GB}|}{|\mathbf{b}_A||\mathbf{n}_{GB}|} = \cos\zeta, \tag{3.3}$$

where ζ is the angle between the incoming slip direction $\mathbf{b}_A$ and the unit normal vectors of GB plane $\mathbf{n}_{GB}$ (see Fig. 3). $E_{release} \sim \tau_{eff}{}^2$ is inversely proportional to the value of p, which agrees that the critical stress of dislocation transmission decreases with decreasing p (Verdhan and Kapoor, 2015). At $p = 0$ (the Burgers vector parallel to the GB plane), the GB does not hinder the dislocation motion.

3.2. Residual dislocation tensor

It has been proposed that the smaller the magnitude of the Burgers vector of residual dislocation $|\triangle\mathbf{b}| = |\mathbf{b}_A - \mathbf{b}_B|$ is (subscripts A and B denote the grains A and B, respectively), the easier the dislocation transmission is (Abuzaid et al., 2012; Gurtin, 2008; Han et al., 2018; Lee et al., 1989; 1990; Li et al., 2009; Schäfer et al., 2016). However, in some cases the magnitude of $\triangle\mathbf{b}$ is insufficient to assess the dislocation transmission behaviors. For example, Shen et al. (1986) found that the angle between the line of intersection of slip planes in grains A and B and the GB plane should be as small as possible for dislocation transmission, meaning that the misorientation of slip planes between grains A and B provides an energy barrier for dislocation transmission (Hémery et al., 2018). To establish the theoretical description of that energy barrier for dislocation transmission, here the dislocation transmissions are categorized into three types as follows and illustrated in Fig. 3.

(1) At $\triangle\mathbf{b} = 0$ and $\mathbf{n}_A \times \mathbf{n}_B = 0$, the leading dislocation can directly transfer to the adjacent grain with zero energy barrier ($E_{barrier} = 0$).
(2) In the case of $\triangle\mathbf{b} \neq 0$ or $\mathbf{n}_A \times \mathbf{n}_B \neq 0$, the dislocations on both sides of a GB are incompatible. Thus the GB would prohibit the dislocation transmission. The dislocation transmits across the GB in a manner of GB absorbing an incoming dislocation followed by emitting an outgoing dislocation (Han et al., 2018).
(3) At $\mathbf{b}_A \cdot \mathbf{n}_{GB} = 0$, the dislocations do not transmit across the GB, and $E_{barrier} = 0$.

Here $\mathbf{b}$ and $\mathbf{n}$ represent the Burgers vector and the unit normal vectors of slip plane, respectively, and can be determined by coordinate transformation matrices from crystal coordinate system to sample frames (see Appendix). The subscripts A, B and GB refer to quantities pertaining to the grains A, B and GB, respectively. One can find that the dislocation transmission is closely associated with spatial relations between the incoming slip system, outgoing slip system and GB plane, and these relations play essential roles in quantitative studies of the critical stress of dislocation transmission.

An edge dislocation is considered as an extra half-plane of atoms [Fig. 3(b)], which can further be characterized by the normal directions of atomic plane ($\mathbf{b}_A$ and $\mathbf{b}_B$) and plane-in vectors ($\mathbf{n}_A$ and $\mathbf{n}_B$) (Hull et al., 2011). The difference between dislocations on incoming and outgoing slip systems can also be characterized by these four vectors. A residual dislocation tensor $\triangle\mathbf{b}_{eff}$ can be obtained to satisfy the conservation law of lattice defect during the dislocation transmission (symbol $\otimes$ denoting the tensor product) as

$$\triangle\mathbf{b}_{eff} = \mathbf{b}_A \otimes (\mathbf{T} \cdot \mathbf{n}_A) - \mathbf{b}_B \otimes \mathbf{n}_B, \tag{3.4}$$

where the transformation tensor $\mathbf{T}$ satisfying the relation $\mathbf{T} \cdot \mathbf{b}_A = \mathbf{b}_B$ with $\mathbf{T}$ is determined through Eq. (A.4) in Appendix. The residual dislocation tensor completely describes the information of slip systems, including the slip direction and slip plane. Notably, here $\mathbf{b}_A \otimes (\mathbf{T} \cdot \mathbf{n}_A)$ is used to represent the incoming dislocation rather than $\mathbf{b}_A \otimes \mathbf{n}_A$, because $\mathbf{n}_A$ and $\mathbf{b}_A$ are not independent (Fig. 4) during the coordinate transformation. The details on their differences are further discussed in Fig. 5.

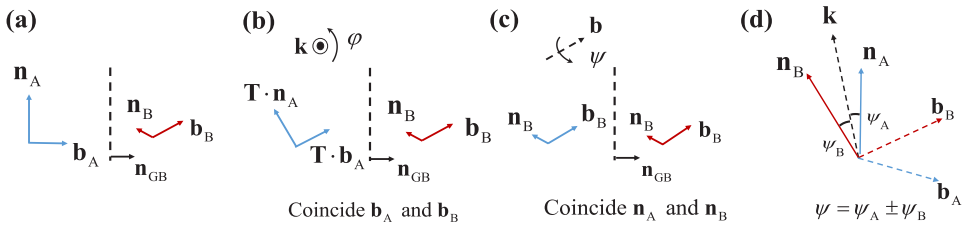

Fig. 4. Illustration of the conversion of dislocation between the incoming slip system and the outgoing slip system. (a)-(c) Convert incoming dislocation to outgoing dislocation, (d) the included angle ψ of $\mathbf{T} \cdot \mathbf{n}_A$ and $\mathbf{n}_B$.

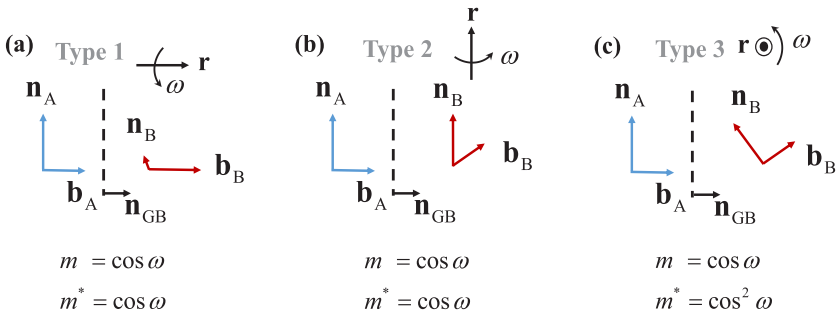

Fig. 5. The plastic compatibility factors m and m^* under three types of misorientations between slip systems in grains A and B.

The magnitude of the residual dislocation tensor is defined as the Frobenius norm of $\triangle\mathbf{b}_{\text{eff}}$, that is

$$\triangle b_{\text{eff}} = \|\triangle\mathbf{b}_{\text{eff}}\| = \sqrt{2b^2 - 2(\mathbf{b}_A \cdot \mathbf{b}_B)(\mathbf{n}_B \cdot \mathbf{T} \cdot \mathbf{n}_A)}. \tag{3.5}$$

Here $\mathbf{n}_B \cdot \mathbf{T} \cdot \mathbf{n}_A$ represents the cosine of angle between $\mathbf{n}_A$ and $\mathbf{n}_B$ when $\mathbf{b}_A$ coincides with $\mathbf{b}_B$ after coordinate transformation [Fig. 4 (b)]. Because $\mathbf{n}_A$ and $\mathbf{n}_B$, respectively, are perpendicular to $\mathbf{b}_A$ and $\mathbf{b}_B$, $\mathbf{n}_B \cdot \mathbf{T} \cdot \mathbf{n}_A$ can be expressed by

$$\mathbf{n}_B \cdot \mathbf{T} \cdot \mathbf{n}_A = \cos(\psi_A \pm \psi_B) = \cos\psi, \tag{3.6}$$

where ψ_A and ψ_B, respectively, are angles between $\mathbf{n}_A$ and $\mathbf{n}_B$ with $\mathbf{k}$, and $\mathbf{k}$ is the axis of the rotation perpendicular to the plane composed by $\mathbf{b}_A$ and $\mathbf{b}_B$. The addition or subtraction of ther these two angles depends on the directions of slip systems [Eq. (3.7)]. The geometric parameters are determined as follows ($\langle\rangle$ representing the angle between vectors)

$$\begin{aligned} \zeta &= \langle \mathbf{b}_A, \mathbf{n}_{GB} \rangle, & \zeta &\in [0, \pi/2], \\ \varphi &= \langle \mathbf{b}_A, \mathbf{b}_B \rangle, & \varphi &\in [0, \pi/2], \\ \psi &= |\langle \mathbf{k}, \mathbf{n}_A \rangle - \text{sgn}[(\mathbf{n}_A \times \mathbf{k}) \cdot (\mathbf{n}_B \times \mathbf{k})]\langle \mathbf{k}, \mathbf{n}_B \rangle|, & \psi &\in [0, \pi/2], \end{aligned} \tag{3.7}$$

and

$$\mathbf{k} = \begin{cases} \dfrac{\mathbf{b}_A \times \mathbf{b}_B}{|\mathbf{b}_A| \cdot |\mathbf{b}_B|}, & \mathbf{b}_A \times \mathbf{b}_B \neq 0, \\ \mathbf{n}_A, & \mathbf{b}_A \times \mathbf{b}_B = 0. \end{cases} \tag{3.8}$$

In comparison with residual Burgers vector $\triangle\mathbf{b} = \mathbf{b}_A - \mathbf{b}_B$, the residual dislocation tensor $\triangle\mathbf{b}_{\text{eff}}$ takes into account both the incompatibility of slip directions and slip planes between two grains. Considering the energy per unit length of dislocation, the energy of residual dislocation E_{rd} is given as (Hull et al., 2011)

$$E_{\text{rd}} = \alpha\mu\triangle b_{\text{eff}}{}^2 = 2\alpha\mu b^2(1 - \cos\varphi\cos\psi). \tag{3.9}$$

where α is the material constant, and φ and ψ are the geometric parameters of slip systems given in Eq. (3.7). E_{rd} describes the energy increase in the system during the dislocation transmission and the term $\cos\varphi\cos\psi$ has the form familiar to the classical plastic compatibility factor proposed by Luster and Morris (1995). At $\mathbf{b}_A \cdot \mathbf{b}_B = 0$ or $\mathbf{n}_A \cdot \mathbf{n}_B = 0$ (i.e. $\cos\varphi\cos\psi = 0$), E_{rd} reaches the maximum value $2\alpha\mu b^2$, meaning that the incoming dislocation is fully absorbed into the GB and a new outgoing dislocation is emitted. At $\psi = 0$ ($\triangle b_{\text{eff}} = |\triangle\mathbf{b}|$), Eq. (3.9) reduces to the model proposed by Li et al. (2009).

To further explore the relation between the magnitude of residual dislocation $\triangle b_{\text{eff}}$ and plastic compatibility factor (Luster and Morris, 1995), Eq. (3.5) can be rewritten with the help of Eq. (3.6) as

$$\triangle b_{\text{eff}} = b\sqrt{2 - 2\cos\varphi\cos\psi} = b\sqrt{2(1 - m)}. \tag{3.10}$$

The factor $m = \cos\varphi\cos\psi$ has a form familiar to the phenomenological plastic compatibility factor $m^* = \cos\varphi\cos\kappa$ with $\kappa = \langle \mathbf{n}_A, \mathbf{n}_B \rangle$ introduced by Luster and Morris (1995). The larger m is, the better plastic compatibility two adjacent grains

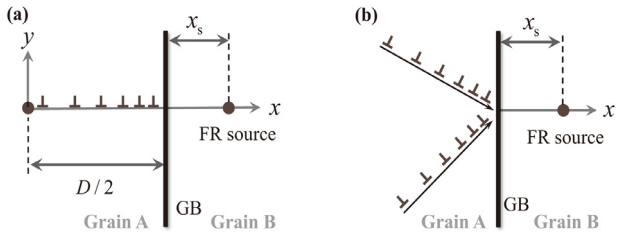

Fig. 6. Schematic diagram of dislocation source activation of (a) single slip dislocation pile-up and (b) multiple slip dislocation pile-ups.

have. Several cases of dislocation mismatch are depicted in Fig. 5 to compare these two factors. It is shown that there are no differences between m and m^* in types 1 and 2 (when the grain B rotates around $\mathbf{b}_A$ or $\mathbf{n}_A$) [Fig. 5(b)]. While the rotation axis switches to the dislocation line ($\mathbf{n}_A \times \mathbf{b}_A$) [Fig. 5(c)], $m^* = \cos^2\omega$ is different from $m = \cos\omega$. In fact, the magnitude of residual Burgers vector in the type 3 is $b\sqrt{2(1-\cos\omega)}$ (Li et al., 2009; Sangid et al., 2011; Wang et al., 2014), which is equal to $\triangle b_{\text{eff}}$ with $m = \cos\omega$. It indicates that the magnitude of residual dislocation tensor [Eq. (3.10)] successfully captures the underlying physics of residual dislocation during dislocation transmission. Also, it is believed that the factor m proposed in this study is more suitable to describe the plastic compatibility than m^*.

3.3. Critical stress for dislocation transmission

In addition to the energy of residual dislocation E_{rd}, GB energy E_{GB} plays an important role in regulating the energy barrier for dislocation transmission E_{barrier}. DD simulations by Li et al. (2009) suggest that E_{barrier} is related to $E_{\text{GB}}b + E_{\text{rd}}$, which describes energy required for the generation of residual dislocation within GBs. Meanwhile, MD simulations (Adlakha and Solanki, 2016; Chandra et al., 2016; Sangid et al., 2011) show that the stability of GB structures is also important in regulating E_{barrier}. An unstable GB structure with high E_{GB} exhibits a low E_{barrier} for dislocation transmission, which obeys an inverse power law relation

$$E_{\text{barrier}} \propto E_{\text{GB}}^{-\beta}, \tag{3.11}$$

where β is the model parameter associated with the materials (Chandra et al., 2016). This relation describes the relationship between the stability of GB structure and the energy barrier for generating the residual dislocation within GBs. Based on the results of these microscopic simulations (Li et al., 2009; Sangid et al., 2011), E_{barrier} can be estimated by

$$E_{\text{barrier}} = \frac{\xi\,(E_{\text{GB}}b + \alpha\mu\triangle b_{\text{eff}}{}^2)}{E_{\text{GB}}^{\beta}}, \tag{3.12}$$

where ξ is the material parameter associated with the GB energy. With Eq. (3.12), it can be deduced that a stable GB structure of low GB energy has a higher energy barrier for absorbing and emitting dislocation than the unstable GB structure of high GB energy. Combining Eqs. (3.1), (3.2) and (3.12), the criterion for the dislocation transmission is obtained as

$$\frac{\pi\,(1-\nu)D\tau_{\text{eff}}{}^2 b}{4\mu\cos\zeta} - \frac{\xi\,(E_{\text{GB}}b + \alpha\mu\triangle b_{\text{eff}}{}^2)}{E_{\text{GB}}^{\beta}} \geq 0. \tag{3.13}$$

When E_{release} of the dislocation pile-up is equal to E_{barrier} of dislocation transmission, the critical resolved shear stress (CRSS) τ_{ct} acting on incoming slip system for the dislocation transmission can be determined by

$$\tau_{\text{ct}} = \tau_0 + \sqrt{\frac{4\mu\xi[E_{\text{GB}}b + 2\alpha\mu b^2(1-\cos\varphi\cos\psi)]\cos\zeta}{\pi b(1-\nu)DE_{\text{GB}}^{\beta}}}. \tag{3.14}$$

The critical stress of dislocation transmission τ_{ct} in Eq. (3.14) is determined based on energy analysis of dislocation transmission, and incorporates the misorientation of slip systems, GB energy E_{GB} and grain size D.

4. Activation of dislocation source in adjacent grains

In addition to the dislocation transmission, dislocation pile-ups can cause the activation of the dislocation source in adjacent grains, which assists the formation of plastic zone and cracks in the region near the GB (Genée et al., 2017). In this section, we investigate the activation of dislocation source in the cases of single or multiple slip dislocation pile-ups.

4.1. Stress fields on outgoing slip system

As illustrated in Fig. 6, FR source locates in adjacent grains at a distance $x_s = x - D/2$ away from the GB. According to Eq. (2.7), the shear stress arising from dislocation pile-ups in the position of FR source can be rewritten as

$$\tau_A^i(x_s, 0) = \tau_{\text{eff}}\left[\frac{x_s^i + D/2}{\sqrt{\left(x_s^i + D\right)x_s^i}} - 1\right], \tag{4.1}$$

where $x_s^i = x_s/p$ (the superscript i corresponds the incoming slip system) and p is the dislocation-GB interaction coefficient defined in Eq. (3.3). At $p = 0$ ($\mathbf{b}_A \cdot \mathbf{n}_{GB} = 0$), we have $\tau_A^i = 0$ as the GB does not hinder the dislocation motion.

Eq. (4.1) gives the shear stress at the coordinate system of incoming slip system in grain A, and the resolved shear stress τ_B^j on the outgoing slip system j in grain B is determined by Livingston and Chalmers (1957)

$$\tau_B^j = [(\mathbf{s}_A \cdot \mathbf{s}_B)(\mathbf{n}_A \cdot \mathbf{n}_B) + (\mathbf{n}_B \cdot \mathbf{s}_A)(\mathbf{s}_B \cdot \mathbf{n}_A)]\tau_A^i, \tag{4.2}$$

in which $\mathbf{s}_A = \mathbf{b}_A/b$ and $\mathbf{s}_B = \mathbf{b}_B/b$ are the unit vectors of the slip directions of incoming and outgoing slip systems, respectively.

It should be noted that that Eq. (4.2) only considers the case of single slip dislocation pile-up and other slip systems are ignored, which is not consistent with the actual physical processes. In fact, the calculation of stress states on FR source should take into account the contribution of all activated slip systems [Fig. 6(b)], such as the face-centered cubic (FCC) materials with 12 slip systems and the body-centered cubic (BCC) materials with 24 slip systems (Hull et al., 2011) Here a more realistic model is proposed that the stress acting on the FR source is obtained by the superposition of the stress fields of all activated slip systems

$$\tau_B^{\beta_B} = \sum_{\alpha_A=1}^{N}\left(\tau_A^{\alpha_A} M_{\beta_B}^{\alpha_A}\right) \tag{4.3}$$

with the transformation coefficient $M_{\beta_B}^{\alpha_A}$ (e.g. 144 components for FCC materials)

$$M_{\beta_B}^{\alpha_A} = (\mathbf{s}_{\alpha_A} \cdot \mathbf{s}_{\beta_B})(\mathbf{n}_{\alpha_A} \cdot \mathbf{n}_{\beta_B}) + (\mathbf{n}_{\beta_B} \cdot \mathbf{s}_{\alpha_A})(\mathbf{s}_{\beta_B} \cdot \mathbf{n}_{\alpha_A}). \tag{4.4}$$

Here N is the number of slip systems, and α_A and β_B represent the slip systems in grains A and B, respectively. $\tau_A^{\alpha_A}$ is the stress produced by the dislocation pile-up at the coordinate system of incoming slip system α_A, and is determined by Eq. (4.1) as

$$\tau_A^{\alpha_A} = \tau_{\text{eff}}^{\alpha_A}\left[\frac{x_s^{\alpha_A} + D/2}{\sqrt{\left(x_s^{\alpha_A} + D\right)x_s^{\alpha_A}}} - 1\right], \tag{4.5}$$

where $\tau_{\text{eff}}^{\alpha_A} = \max(|\boldsymbol{\sigma}:\mathbf{R}^{\alpha_A}| - \tau_0, 0)/\text{sgn}(\boldsymbol{\sigma}:\mathbf{R}^{\alpha_A})$ is the effective stress on the slip system α_A, in which $\boldsymbol{\sigma}$ is the Cauchy stress tensor and $\mathbf{R}^{\alpha_A}$ is the Schmid tensor defined by $\mathbf{R}^{\alpha_A} = \left(\mathbf{s}_{\alpha_A} \otimes \mathbf{n}_{\alpha_A} + \mathbf{n}_{\alpha_A} \otimes \mathbf{s}_{\alpha_A}\right)/2$.

Substituting Eq. (4.5) into Eq. (4.3), the shear stress at the location of FR sources under multiple slip dislocation pile-ups is obtained as

$$\tau_B^{\beta_B} = \sum_{\alpha_A=1}^{N}\left[\tau_{\text{eff}}^{\alpha_A} M_{\beta_B}^{\alpha_A}\frac{x_s^{\alpha_A} + D/2 - \sqrt{\left(x_s^{\alpha_A} + D\right)x_s^{\alpha_A}}}{\sqrt{\left(x_s^{\alpha_A} + D\right)x_s^{\alpha_A}}}\right]. \tag{4.6}$$

In the derivation of Eq. (4.6), we adopt the hypothesis proposed by Davis et al. (1966) that the stress acting on the dislocation source is little affected by the parallel slip systems. Hence, the influence of the parallel slip system could be ignored without losing the effectiveness of $\tau_B^{\beta_B}$ given in Eq. (4.6).

4.2. Critical stress for activating Frank-Read sources

Under certain shear stress, the FR dislocation source in metallic materials can continuously emit dislocations by bowing out dislocation segments to form a dislocation loop (Zhu et al., 2014). Once the FR source is activated, it continuously emits a series of dislocations. According to the bowing-dislocation model proposed by Foreman (1967), the critical activation stress of FR source can be expressed as

$$\tau_{FR} = C_0\frac{\mu b}{2\pi l_s}\ln\frac{l_s}{r_0}, \tag{4.7}$$

where τ_{FR} is the local stress required for activating FR sources, C_0 is the coefficient related to the character of dislocation segments and Poisson ratio v ($C_0 = 1$ for edge dislocation and $v \approx 1/3$), l_s is the length of dislocation segment in stress-free state, and r_0 is the radius of the dislocation core.

Table 1
Predictions of outgoing slip systems for dislocation-GB interactions.

N	Rotation axis/angle	[110]/70.5°		[110]/70.5°		$[\bar{1}1\bar{1}]/60°$		[121]/180°		[101]/-40°	
	Slip system	$\triangle b_{\text{eff}}/b$	Exp[a]	$\triangle b_{\text{eff}}/b$	Exp[a]	$\triangle b_{\text{eff}}/b$	Exp[a]	$\triangle b_{\text{eff}}/b$	Exp[a]	$\triangle b_{\text{eff}}/b$	Exp[b]
1	$[1\bar{1}0](111)$	1.07	-	0.66	A	1.15	-	1.11	-	1.22	-
2	$[10\bar{1}](111)$	1.00	-	1.11	I	1.11	-	0.67	I/A	1.06	-
3	$[01\bar{1}](111)$	1.41	-	1.15	-	0.66	A	1.11	-	1.33	-
4	$[110](\bar{1}11)$	1.00	-	1.07	-	1.41	-	1.00	-	1.41	-
5	$[101](\bar{1}11)$	1.07	-	0.97	B	1.07	-	1.41	-	1.00	-
6	$[01\bar{1}](\bar{1}11)$	1.41	-	1.33	-	0.97	-	1.07	-	1.06	-
7	$[110](1\bar{1}1)$	1.29	-	1.11	-	1.41	-	1.29	-	1.41	-
8	$[10\bar{1}](1\bar{1}1)$	1.29	-	1.29	-	1.29	-	1.15	-	0.6	A
9	$[011](1\bar{1}1)$	1.16	-	1.41	-	1.11	-	1.29	-	0.68	I
10	$[1\bar{1}0](11\bar{1})$	1.11	-	0.97	-	1.33	-	1.07	-	1.34	-
11	$[101](11\bar{1})$	1.11	I	1.07	-	0.97	I	1.41	-	1.28	-
12	$[011](11\bar{1})$	0.67	A	1.41	-	1.07	-	1.00	-	1.4	-
	Normal of GB	$(\bar{1}11)$		$(\bar{1}11)$		$(\bar{1}11)$		$(1\bar{1}1)$		$(1\bar{1}1)$	

I, A and B represent the incoming slip system, outgoing primary and secondary slip systems, respectively. Superscripts *a* and *b* represent the experimental results adopted from Lee et al. (1989) and Cui et al. (2014), respectively.

When the local stress in the position of FR source reaches τ_{FR}, the dislocation source can be activated. Combining Eqs. (4.1), (4.2) and (4.7), the critical shear stress τ_{cs} for the single slip system can be written as

$$\tau_{\text{cs}} = \tau_0 + \frac{\sqrt{\left(x_{\text{s}}^i + D\right)x_{\text{s}}^i}}{x_{\text{s}}^i + D/2 - \sqrt{\left(x_{\text{s}}^i + D\right)x_{\text{s}}^i}}\left[(\mathbf{s}_{\text{A}}\cdot\mathbf{s}_{\text{B}})(\mathbf{n}_{\text{A}}\cdot\mathbf{n}_{\text{B}}) + (\mathbf{n}_{\text{B}}\cdot\mathbf{s}_{\text{A}})(\mathbf{s}_{\text{B}}\cdot\mathbf{n}_{\text{A}})\right]^{-1}\frac{\mu b}{2\pi l_{\text{s}}}\ln\frac{l_{\text{s}}}{r_0}. \tag{4.8}$$

For multiple slip systems, the critical applied stress $\boldsymbol{\sigma}_{\text{cs}}$ for activation of FR source on the slip system β_{B} can be determined by combining Eqs. (4.6) and (4.7) as

$$\sum_{\alpha_{\text{A}}=1}^{N}\left[\frac{\max\left(|\sigma:\mathbf{R}^{\alpha_{\text{A}}}| - \tau_0, 0\right)}{\text{sgn}(\sigma:\mathbf{R}^{\alpha_{\text{A}}})}\cdot M_{\beta_{\text{B}}}^{\alpha_{\text{A}}}\frac{x_{\text{s}}^{\alpha_{\text{A}}} + D/2 - \sqrt{\left(x_{\text{s}}^{\alpha_{\text{A}}} + D\right)x_{\text{s}}^{\alpha_{\text{A}}}}}{\sqrt{\left(x_{\text{s}}^{\alpha_{\text{A}}} + D\right)x_{\text{s}}^{\alpha_{\text{A}}}}}\right] - \frac{\mu b}{2\pi l_{\text{s}}}\ln\frac{l_{\text{s}}}{r_0} = 0. \tag{4.9}$$

Note that Eq. (4.9) gives σ_{cs} for activating FR source in grain B. To facilitate analysis, the critical activation stress is defined as the minimum of all critical applied stresses. In addition, in Eqs. (4.8) and (4.9), the RSS of external applied stress on grain B has been ignored because it is relatively small compared with the stress concentration ahead of the dislocation pile-up (Pan et al., 2015). As the stress caused by dislocation pile-ups increases, the activation of FR sources becomes easier, consistent with the stress criterion proposed by Shen et al. (1986).

5. Numerical results of dislocation transmission

The developed theoretical model is employed to study the dislocation transmission behaviors. Firstly, the outgoing slip systems are predicted and compared with experimental results of stainless steels (in Section 5.1) (Cui et al., 2014; Lee et al., 1989). Secondly, the effect of GB misorientation angle θ on critical transmission stress τ_{ct} is discussed in Section 5.2. Finally, τ_{ct} displays a dependence on grain size D in Section 5.3.

5.1. Predictions of outgoing slip systems

Taking FCC materials as an example, the critical transmission stress τ_{ct} [Eq. (3.14)] of the preferential outgoing slip system corresponds to the smallest τ_{ct} among the 12 slip systems. Then it can also be inferred that the secondary outgoing slip system has the second smallest τ_{ct}. Accordingly, activation behaviors of outgoing slip systems can be predicted by comparing the corresponding τ_{ct}.

Theoretical predictions [Eq. (3.14)] are compared with the experimental data of 310 stainless steel (Lee et al., 1989) and 304 stainless steel (Cui et al., 2014). According to Eqs. (3.12) and (3.14), the critical transmission stress τ_{ct} for a fixed GB is only associated with the magnitude of residual dislocation tensor $\triangle b_{\text{eff}}$. Thus the factor $\triangle b_{\text{eff}}/b$ can be used to evaluate the activation behaviors of outgoing slip systems, i.e. the smaller the factor $\triangle b_{\text{eff}}/b$ is, the easier the dislocation transmission becomes.

Experimental results and theoretical predictions of five groups of dislocation transmissions are presented in Table 1. It can be seen that the slip system with the smallest value of $\triangle b_{\text{eff}}/b$ is activated preferentially. Moreover, in the second group experimental data, the activation of the secondary outgoing slip system has the second smallest $\triangle b_{\text{eff}}/b$, which verifies the validity of our model. Meanwhile, in the fifth group of the experimental data, Cui et al. (2014) found that the outgoing slip

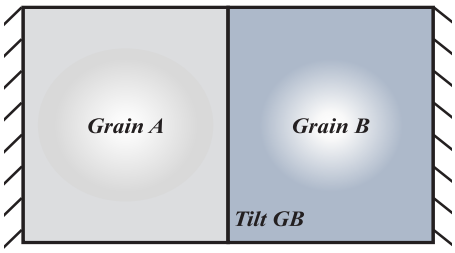

Fig. 7. Schematic of a bi-crystal that GBs on both sides are assumed to be impenetrable.

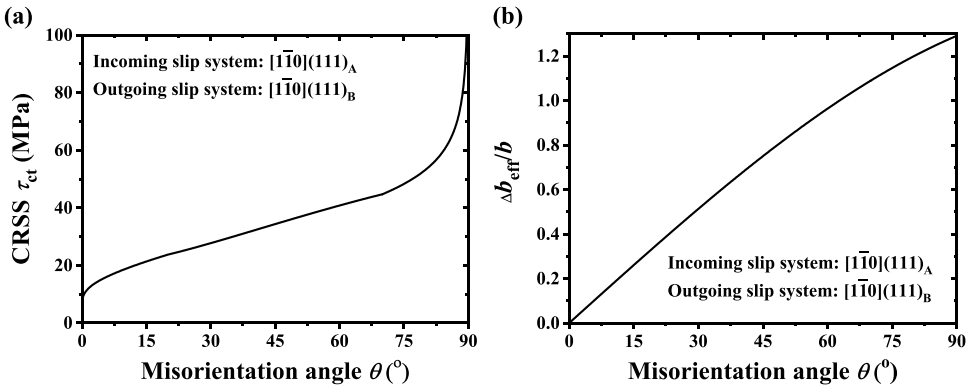

Fig. 8. Variations of τ_{ct} and $\triangle b_{eff}/b$ with θ for slip system $[1\bar{1}0](111)$. Subscripts A and B represent the corresponding grains.

system $[011](11\bar{1})$ with high $\triangle b_{eff}/b$ is activated. It seems to be contradicting with our conclusions. However, the significant stress concentration induced by dislocation pile-up was observed on this slip system (Cui et al., 2014), which means the possible activations of dislocation source.

In most previous studies, the outgoing slip system was predicted by LRB criteria (Lee et al., 1989; 1990), in which three factors, that is, residual dislocation, resolved shear stress and interaction angle are involved. In comparison with LRB criteria, using $\triangle b_{eff}/b$ as proposed here is more convenient as it is quite difficult to identify which of these three factors in LRB criteria dominates the dislocation transmission.

5.2. Critical stress of dislocation transmission

Quantitative investigation on the dislocation transmission with various misorientations angles θ is achievable according to Eq. (3.14). As depicted in Fig. 7, bi-crystal aluminum (Al) with the $\langle 100\rangle$ tilt GBs in the range of 0° to 90° are selected to investigate the critical stress of dislocation transmission, and material parameters are listed in Table 2.

Based on the experimental data (Hasson and Goux, 1971), the energies of $\langle 100\rangle$ tilt GB have a statistical expression

$$E_{GB} = \begin{cases} k\theta/\theta_1, & \theta \in [0^o, \theta_1], \\ k, & \theta \in (\theta_1, \theta_2), \\ k(90^o - \theta)/(90^o - \theta_2), & \theta \in [\theta_2, 90^o]. \end{cases} \tag{5.1}$$

The HAGB energy k is taken approximately as 600 mJ/m^2 for Al (Hasson and Goux, 1971), and the parameters θ_1 and θ_2 are adopted as 20° and 70°, respectively.

5.2.1. Dislocation transmission for single slip system

Dislocation transmission for single slip system $[1\bar{1}0](111)$ is investigated, and the normal direction of the GB plane is set as $\mathbf{n}_{GB}$=(010). Fig. 8 displays the variation of the critical transmission stress τ_{ct} and $\triangle b_{eff}/b$ with the misorientation angle θ.

At $\theta = 0^o$, we have $\tau_{ct} = \tau_0$ as $\triangle b_{eff}/b = 0$. As θ increases, τ_{ct} increases due to the increased $\triangle b_{eff}$ meaning that HAGBs provide the high resistance for dislocation transmission. With a further increase of θ, τ_{ct} increases rapidly. One reason is the decreased GB energy after $\theta > 70^o$. Another reason is that the outgoing slip system here is fixed. In fact, in the case of multiple slip systems, while τ_{ct} of one outgoing slip systems is very high, the outgoing slip systems can switch to another one with low $\triangle b_{eff}$. This result indicates that the dislocation transmission behavior is strongly affected by both $\triangle b_{eff}/b$ and E_{GB}. A high energy barrier of dislocation transmission can be provided by the low energy GB, whose interfacial energy is affected by misorientation angle, GB plane normal and free volume (Olmsted et al., 2009). Combined with our results, one can conclude that for the HAGB with low GB energy such as $\Sigma 3$ coherent GB, E_{GB} would be the main factor affecting the

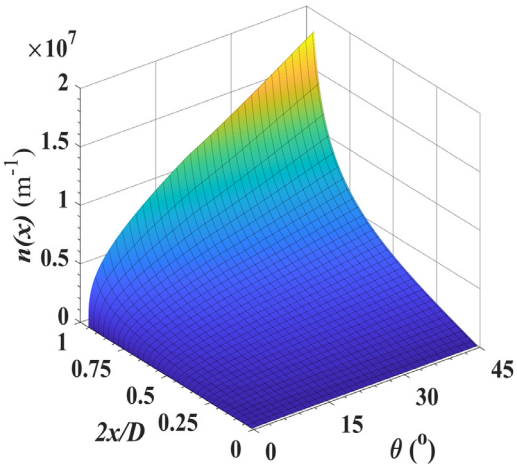

Fig. 9. Dislocation linear density $n(x)$ as a function of the dislocation position x and misorientation angle θ.

dislocation transmission (Sangid et al., 2011). Whereas for the general HAGBs, the GB misorientation angle usually plays the dominant role in affecting dislocation transmission, as presented in Fig. 8.

Combining Eqs. (2.3) and (3.14), the dislocation linear density $n(x)$ for dislocation transmission with various θ is shown in Fig. 9. Here, we only discuss the case at $\theta < 45^{\circ}$ as the tilt GBs is symmetric about $\theta = 45^{\circ}$. As shown in Fig. 9, $n(x)$ in the vicinity of $x \approx 0$ is close to 0 m^{-1}. Due to the dislocation pile-up formation, $n(x)$ increases as x increases until a maximum value is reached in the vicinity of the GB. $n(x)$ increases with increasing θ. At $\theta \approx 45^{\circ}$, the dislocation density is approximately 1.8×10^7 m^{-1} at $x \approx 250$ nm, comparable to the experimental result 2.4×10^7 m^{-1} ($x \approx 250$ nm) measured by Shen et al. (1986). As θ decreases, the energy barrier for dislocation transmission is reduced, and the dislocation pile-up relaxes.

5.2.2. Dislocation transmission for multiple slip systems

The dislocation transmission in multiple slip systems is investigated here. For FCC Al, there are 144 components of dislocation transmissions for 12 incoming and outgoing slip systems. For simplicity without losing generality, here we only consider uniaxial loading. According to Eq. (3.14), the external applied stress of dislocation transmission σ_{ct} under uniaxial loading can be written as

$$\sigma_{\mathrm{ct}} = \min_{\substack{\beta_{\mathrm{B}} = 1,12 \\ \alpha_{\mathrm{A}} = 1,12}} \left[\frac{\tau_0}{R^{\alpha_{\mathrm{A}}}} + \sqrt{\frac{4\mu\xi\left[E_{\mathrm{GB}} b + \alpha\mu\left(\triangle b_{\mathrm{eff}}^{\alpha_{\mathrm{A}},\beta_{\mathrm{B}}}\right)^2\right]\cos\zeta}{\pi b(1-\nu) D E_{\mathrm{GB}}^{\beta} (R^{\alpha_{\mathrm{A}}})^2}} \right], \tag{5.2}$$

where $R^{\alpha_{\mathrm{A}}}$ is the Schmid factor of the slip system α_{A}, $\triangle b_{\mathrm{eff}}^{\alpha_{\mathrm{A}},\beta_{\mathrm{B}}}$ is the magnitude of the residual dislocation tensor for dislocation transmission from α_{A} into β_{B} slip systems. As discussed in Section 3.1, when the slip direction is parallel to the GB plane ($p = 0$), the dislocation transmission across the GB does not occur ($\sigma_{\mathrm{ct}} = 0$). Therefore, $\mathbf{n}_{\mathrm{GB}}$ is set as (012) to avoid the situation of $p = 0$. It is noted that this GB is an asymmetric tilt GB. Here we still adopt Eq. (5.1) to describe the GB energy because there is no significant difference in GB energy between ⟨100⟩ symmetric tilt GB and ⟨100⟩ asymmetric tilt GB (Tschopp and McDowell, 2007).

Fig. 10 shows σ_{ct} as a function of θ under loading directions along [010], [111] and [012] (see Table 2 for model parameters). It is shown that σ_{ct} of LAGB is smaller than that of HAGB, consistent with experimental results (Bieler et al., 2019; Hémery et al., 2018; Joseph et al., 2018). In addition, the profile $\sigma_{\mathrm{ct}}(\theta)$ exhibits the symmetry with respect to $\theta = 45^{\circ}$ due to the symmetry of tilt GB. At the stage of $\theta < 45^{\circ}$, the evolutions of σ_{ct} have the same trend as a result of single slip in Fig. 8(a). Contrary to the rapid increase of σ_{ct} at $\theta > 45^{\circ}$ in Fig. 8(a), σ_{ct} decreases with increasing θ. This is mainly due to the switching of the incoming and outgoing slip system to ones corresponding to the lowest σ_{ct} or $\triangle b_{\mathrm{eff}}/b$.

According to previous studies (Ding et al., 2016; Guo et al., 2014; Hémery et al., 2018; Joseph et al., 2018), there is no essential difference in the physical mechanism of GB-dislocation interaction between HCP and other crystalline materials. Thus, the developed model can be generalized to study the dislocation transmission behavior of the hexagonal-close-packed (HCP) crystalline materials by taking into account the anisotropy of the slip systems. For HCP materials, the operation of slip systems is associated with the grain size and strain rate (Li et al., 2018)

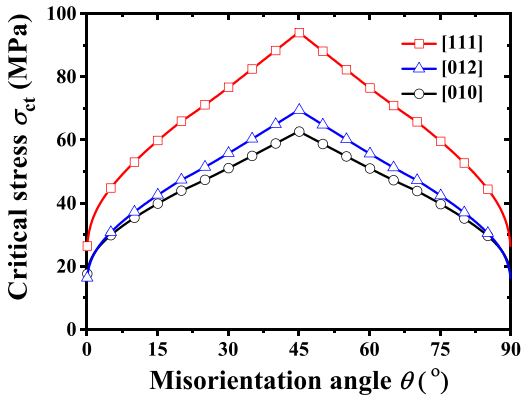

Fig. 10. Variation of σ_{ct} with θ under loading of different directions [111], [012] and [010].

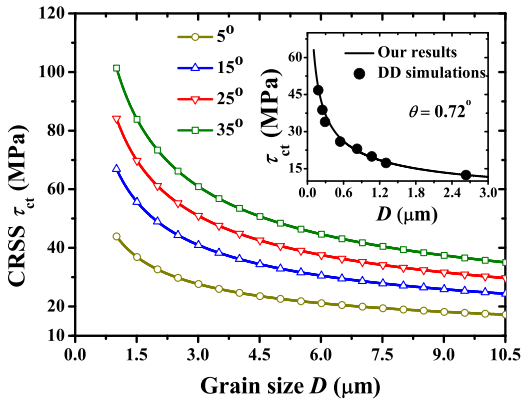

Fig. 11. Variations of τ_{ct} with D for θ of 5°, 15°, 25° and 35°. The result of DD simulation is adopted from Kapoor and Verdhan (2017).

5.3. Hall-Petch effect at different misorientation angles

It is well known that the yield stress of polycrystalline materials displays the Hall-Petch effect (Hall, 1951; Li et al., 2009; Petch, 1953), i.e. the yield stress σ_y increases with decreasing grain size D ($\sigma_y \propto D^{-0.5}$). In this study, such an effect can be further interpreted by the decreasing energy release rate with decreasing D [Eq. (2.6)]. According to Eq. (3.14), the evolution of $\tau_{ct} \sim D$ at different θ is investigated. Here the incoming slip system is fixed as $[1\bar{1}0](111)$, and $\mathbf{n}_{GB}$ is set as $(1\bar{1}0)$.

τ_{ct} at θ= 5°, 15°, 25° and 35° display the Hall-Petch effect (Fig. 11). As θ decreases, the Hall-Petch effect becomes weaker, consistent with the simulation results by Li et al. (2009) that the Hall-Petch effect with HAGB is more apparent than that with LAGB. As shown in the inset of Fig. 11, our theoretical model shows good agreement with DD simulations (Kapoor and Verdhan, 2017). The LAGB investigated by DD simulations is composed of a series of edge dislocations with spacing of $S = 80b$ (Kapoor and Verdhan, 2017), and its misorientation angle can thus be approximated as $\theta = b/S \approx 0.72°$ (Hirth and Lothe, 1982). As the dislocation motion resistance τ_0 was not considered in DD simulations, $\tau_0 = 0$ MPa is taken in our theoretical analysis.

6. Numerical results of dislocation source activation

The dislocation pile-up induced dislocation source activations in adjacent grains is investigated in this section. Using Eq. (2.7), the shear stress field τ_{xy}/τ_{eff} ahead of the dislocation pile-up is plotted in Fig. 12 as a function of $x_{GB} = x - D/2$. The stress concentration exists in the region near the GB and gradually vanishes as x_{GB} increases, verified by experiments (Britton and Wilkinson, 2012; Guo et al., 2014). Under the stress concentration, dislocation sources with activation stress less than local stress can be activated (Cottrell, 1958; Joseph et al., 2018). Here the bi-crystal Al (see Fig. 7) is taken as an example for investigation, and tilt GBs with θ from 0° to 90° are considered. According to the experimental observations (Genée et al., 2017; Joseph et al., 2018), the distance from the dislocation nucleation site or the core of plastic zone to GBs falls in a range from around several hundred nanometers to several micrometers, and the average grain size ranges from around 4 μm to 350 μm. Therefore, we adopt the source distance of 1 μm for a grain size of 10 μm, as listed in Table 2.

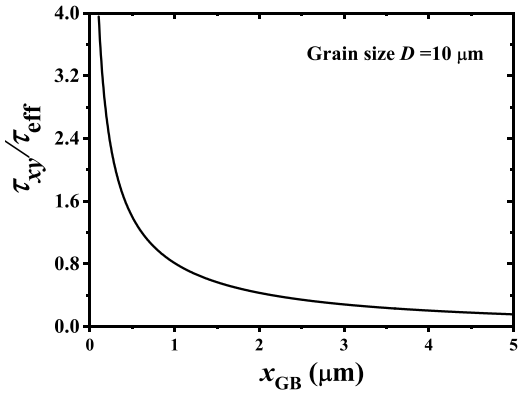

Fig. 12. Normalized shear stress $\tau_{xy}/\tau_{\text{eff}}$ ahead of a dislocation pile-up as a function of the distance $x_{\text{GB}} = x - D/2$ from the GB.

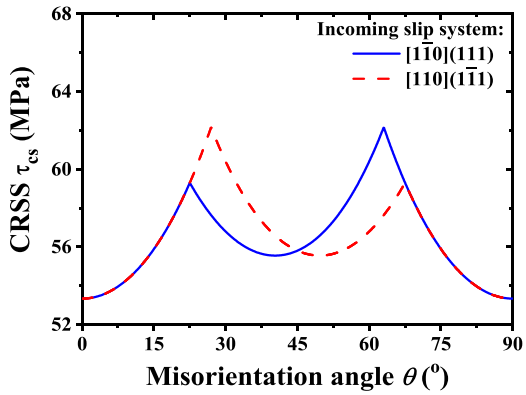

Fig. 13. Variation of τ_{cs} with θ of single slip dislocation pile-up.

6.1. Critical activation stress of single slip system

For the single slip dislocation pile-up, two case studies are performed with different fixed incoming slip systems $[1\bar{1}0](111)$ and $[110](1\bar{1}1)$, and $\mathbf{n}_{\text{GB}}$=(010). Fig. 13 shows the CRSS τ_{cs} from Eq. (4.8) as a function of the misorientation angle θ for the single pile-up (model parameters listed in Table 2). The lowest τ_{cs} of these two cases has the same tendency as a previous study (Spearot and Sangid, 2014). Owing to the symmetry of tilt GB, the evolution of τ_{cs} for two incoming slip systems is symmetric with respect to $\theta = 45^{\text{o}}$. Therefore, we only need to discuss the evolution of critical stresses with θ from 0° to 45°. At $\theta = 0^{\text{o}}$, the critical stress has the minimum value as the transformation coefficient $M_{\beta_{\text{B}}}^{\alpha_{\text{A}}}$ reaches the maximum value 1 [Eq. (4.4)]. With increasing θ, both the stress on the FR source and $M_{\beta_{\text{B}}}^{\alpha_{\text{A}}}$ decrease, leading to the initial increase of critical stress τ_{cs}. If θ continues to increase, there will be several turning points in the evolution of τ_{cs} due to the activation of FR source in other outgoing slip systems, which leads to the decrease of τ_{cs}. That is analogous to the switching of outgoing slip system as discussed in Section 5.3.

6.2. Critical activation stress of multiple slip systems

Activation behaviors of dislocation source for FCC material Al upon uniaxial loading is studied, and the critical activation stress σ_{cs} given in Eq. (4.9) (uniaxial loading) is the minimum critical activation stresses of 12 potential outgoing slip systems. $\mathbf{n}_{\text{GB}}$ is set as (012) to avoid $\mathbf{b}_{\text{A}} \cdot \mathbf{n}_{\text{GB}} = 0$. The loading directions are selected as [001], [010], [012] and [013] (four columns in Fig. 14) and dislocation motion resistance τ_0 is set as 5.23 MPa, 40 MPa and 100 MPa (three rows in Fig. 14 to investigate these effects on critical activation stress.

Fig. 14 shows that the profile of $\sigma_{\text{cs}}(\theta)$ exhibits feature of sensitivity to loading direction. This sensitivity to loading direction is mainly attributed to the variety of Schmid factors under different loading directions, which results in different RSSs on incoming slip systems, and then indirectly influences dislocation pile-ups and local stress on the FR source. Unlike the case of dislocation transmission which is prone to occur at LAGBs, due to the sensitivity of $\sigma_{\text{cs}}(\theta)$ to the loading direction, it is difficult to access the ease of activation behaviors in polycrystals through θ. On this basis, the reported results of dislocation source activation in polycrystals can be interpreted, in which the activation of FR sources has no obvious tendency with misorientation angles (Genée et al., 2017).

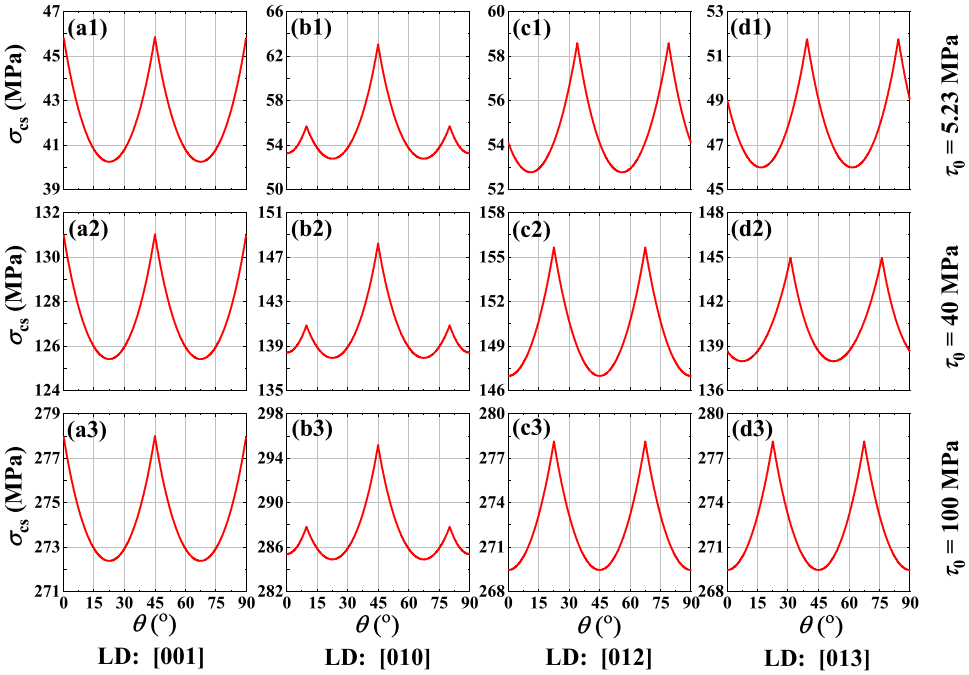

Fig. 14. σ_{cs} versus θ of multiple slip dislocation pile-ups at different τ_0 and loading directions (LDs). Model parameters are given in Table 2.

Table 2
Parameters for model calculation of FCC Al.

Parameters	Definition	Values	Units
μ	Shear modulus	[a]26.1	GPa
b	Magnitude of Burgers vector	[a]0.286	nm
α	Dislocation efficient	[a]0.5	-
ν	Poisson ratio	[a]0.33	-
τ_0	Dislocation motion resistance	[a]5.23	MPa
D	Grain size	10	μm
ξ	Model parameter	0.08	$J^{\beta}/m^{2\beta}$
β	Model parameter	0.4	-
k	HAGB energy	[b]0.6	J/m^2
θ_1	Tilt GB parameter	[b]20	deg
θ_2	Tilt GB parameter	[b]70	deg
l_s	Dislocation segment length	[c]286	nm
r_0	Radius of dislocation core	[c]0.286	nm
x_s	Distance of FR source from GB	1.0	μm

Values with superscripts *a*, *b*, and *c* are adopted from [a] Meyers and Chawla (2009), [b] Hasson and Goux (1971) and [c] Foreman (1967), respectively.

Because the effective stress on incoming slip system is determined by RSS and τ_0, the dislocation pile-up is influenced by τ_0 [Eq. (2.1)]. Under loading directions of [001] and [010] [see Figs. 14(a1)-(a3) and (b1)-(b3)], σ_{cs} increases as τ_0 increases, but the positions θ of local maximum of σ_{cs} remain the same values. In the cases of loading directions of [012] and [013], the increase of τ_0 shifts the $\sigma_{cs}(\theta)$ profile leftward, as shown in Figs. 14(c1)-(c3) and (d1)-(d3). It means that the effect of τ_0 is coupled with the loading direction effects to affect the behavior of dislocation source activation.

In this study, a more realistic theory of dislocation source activation is proposed in which the contribution of all slip systems is taken into account in the calculation of the stress on FR source. The model results are effective to explain some previous confusing experimental phenomena, such as the irregular distribution of misorientation angles for dislocation source activation (Genée et al., 2017).

6.3. Comparison between critical stresses for dislocation transmission and FR source activation

In Sections 5.2 and 6.2, the critical stress for dislocation transmission σ_{ct} and dislocation source activation σ_{cs} are discussed, and the low misorientation angle θ is shown to facilitate the dislocation transmission. In addition to θ, the activation of dislocation source depends on the loading direction. To explore the competition behaviors of these two mechanisms, here

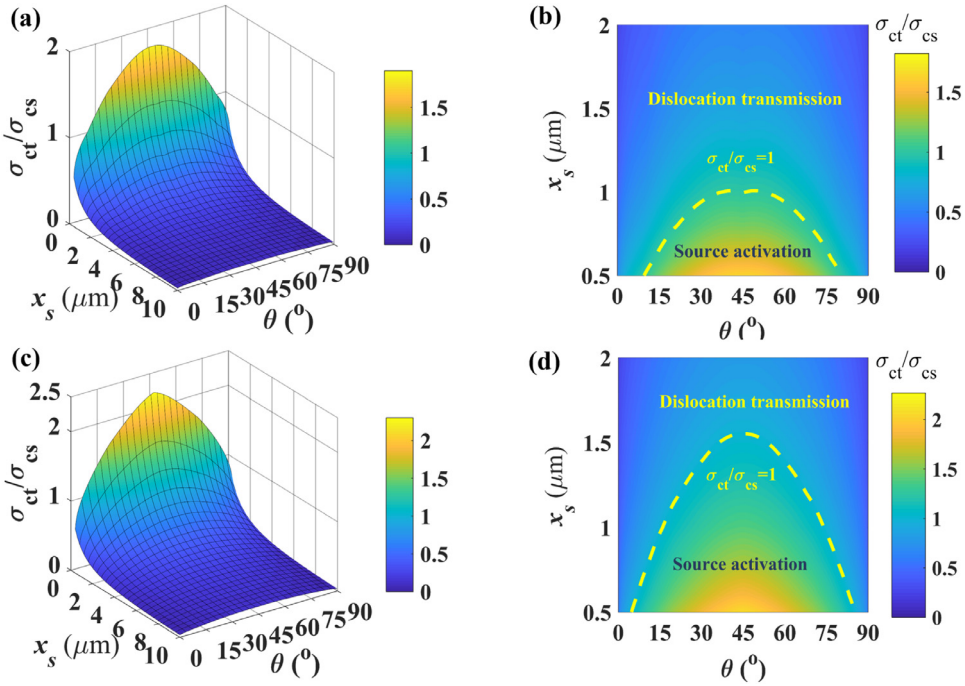

Fig. 15. Critical stress ratio σ_{ct}/σ_{cs} between dislocation transmission and source activation under loading directions (a-b) [010] and (c-d) [001].

we analyze the ratio of critical stresses σ_{ct}/σ_{cs}. σ_{cs} and σ_{ct} are determined from Eqs. (4.9) and (5.2), respectively. Material parameters are listed in Table 2, and $\mathbf{n}_{GB}$=(012). Two loading directions [010] and [001] are selected.

Fig. 15 displays σ_{ct}/σ_{cs} with various θ and x_s. As x_s increases, σ_{ct}/σ_{cs} decreases, consistent with Fig. 12 showing that the stress ahead of the dislocation pile-up decreases with increasing distance from GB. The FR source far away from GB is difficult to be activated, and thus the plastic zone induced by dislocation pile-ups forms near GB as shown in Fig. 2. σ_{ct}/σ_{cs} increases as θ increases from 0º to 45º, meaning that the dislocation transmission is easier to occur at LAGB, but FR source activation is prone to occur at HAGB. The underlying physical reasons are that, the dislocation transmission across LAGB suffers small resistance, and the dislocation pile-up can be relaxed and there would not be enough stress to activate the dislocation source. However, the HAGB provides a high resistance for dislocation transmission, leading to significant dislocation pile-ups and dislocation source activations. It is noted that this conclusion is only valid for ⟨100⟩ tilt GBs. For certain GBs with low interfacial energy, the dislocation-GB interactions could be more affected by E_{GB} than θ (Sangid et al., 2011).

The contour plots of σ_{ct}/σ_{cs} are shown in Figs. 15(b) and (d). At $\sigma_{ct}/\sigma_{cs} > 1$, activation of FR source dominates dislocation-GB interactions. At $\sigma_{ct}/\sigma_{cs} \approx 1$, FR source activation and dislocation transmission can occur simultaneously as observed in experiments (see Fig. 2) (Genée et al., 2017). At $\sigma_{ct}/\sigma_{cs} < 1$, dislocation transmission prevails. In addition, σ_{ct}/σ_{cs} in the loading direction [010] is smaller than that in the loading direction [001], which means that the source activation further away from GB can be activated under the loading direction [001]. Although the FR source activation is sensitive to the loading direction, the activation behavior under different loading directions still does not occur at very small θ due to the relaxation of dislocation pile-ups during dislocation transmission, which is consistent with the experimental observations (Genée et al., 2017; Joseph et al., 2018).

Finally, the relation between three factors proposed in LRB criteria (Lee et al., 1989; 1990) and dislocation-GB interactions can be clarified. Overall, the misorientation angle between the incoming and outgoing slip planes and the residual dislocation can equally affect the energy barrier for dislocation transmission. However, the RSS on the outgoing slip system arising from the dislocation pile-up dominates the activation of dislocation sources in adjacent grains. In this study, the effects of residual stress, local differences in statistical strength, non-uniform inter- and intragranular mechanical fields on the dislocation-GB interaction are not considered, which in certain cases might play collective roles in affecting the dislocation-GB interaction. A thorough and cross-scale theoretical modeling considering these factors is challenging and deserves further dedicated investigations with incorporation of crystal plasticity finite element method.

7. Conclusions

In this work, a theoretical model is developed to study the dislocation-grain boundary (GB) interaction with focus on the dislocation transmission and dislocation source activation. This model not only quantitatively describes the microstructure-

related dislocation-GB interaction, but also clarifies the misorientation angle-dependent competition relation between dislocation transmission and dislocation source activation. Specifically, the energy released from the end of dislocation pile-up goes over the energy barrier characterized by the residual dislocation tensor and GB energy to drive the dislocation transmission across GB. Based on the residual dislocation tensor, the critical transmission stress can be connected with the phenomenological plastic compatibility factor. In addition, critical stress that activates Frank-Read (FR) sources in adjacent grains is examined by evaluating the local stress ahead of the dislocation pile-up. In contrast to the formulation of single slip dislocation pile-up, a more realistic formulation of multiple slip dislocation pile-ups is proposed to investigate the activation behaviors of dislocation source, in which the contribution of all incoming slip systems is taken into account in the calculation of the stress on FR source.

⟨100⟩ tilt GBs are taken as examples to systematically study the dislocation-GB interaction. Main conclusions are summarized as follows.

(1) The quantitative analysis shows that with increasing misorientation angle, the critical stress of dislocation transmission increases, which also displays the dependence of grain size. In addition, the model accurately predicts the activation of primary and secondary outgoing slip systems observed in reported experiments.
(2) Critical stress of dislocation source activation is quantitatively analyzed for single and multiple slip dislocation pile-ups. Compared with single slip system, the result of multiple slip systems shows lower critical activation stress and loading direction sensitivity, interpreting the reported results of dislocation source activation in polycrystals.
(3) Comparing the activation stresses of dislocation transmission and dislocation source activation for ⟨100⟩ tilt GBs, the dislocation transmission dominates dislocation-GB interaction at low angle GB, whereas the dislocation source activation is more likely to occur at high angle GB.

Declaration of Competing Interest

The authors have no conflicts of interest to disclose.

CRediT authorship contribution statement

Wenbin Liu: Conceptualization, Methodology, Software, Validation, Formal analysis, Writing - original draft. **Ying Liu:** Methodology, Writing - review & editing. **Haonan Sui:** Methodology, Writing - review & editing. **Lirong Chen:** Investigation, Writing - review & editing. **Long Yu:** Investigation, Writing - review & editing. **Xin Yi:** Writing - review & editing, Supervision, Project administration, Funding acquisition. **Huiling Duan:** Writing - review & editing, Supervision, Project administration, Funding acquisition.

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Appendix

The coordinate transformation tensor from the crystal coordinate system to sample frames is determined by Euler angles (ψ_1, θ_1, φ_1) as (Bunge definition)

$$\mathbf{g} = g^i_{j'} e_i \otimes e^{j'} \tag{A.1}$$

with the coordinate transformation tensor $\mathbf{g}$ given as

$$\left[g^i_{j'}\right] = \begin{bmatrix} \cos\psi_1\cos\varphi_1 - \sin\psi_1\sin\varphi_1\cos\theta_1 & \sin\psi_1\cos\varphi_1 + \cos\psi_1\sin\varphi_1\cos\theta_1 & \sin\varphi_1\sin\theta_1 \\ -\cos\psi_1\sin\varphi_1 - \sin\psi_1\cos\varphi_1\cos\theta_1 & -\sin\psi_1\sin\varphi_1 + \cos\psi_1\cos\theta_1\cos\varphi_1 & \cos\varphi_1\sin\theta_1 \\ \sin\psi_1\sin\theta_1 & -\cos\psi_1\sin\theta_1 & \cos\theta_1 \end{bmatrix}. \tag{A.2}$$

Here $\mathbf{e}_1 = [100]$, $\mathbf{e}_2 = [010]$ and $\mathbf{e}_3 = [001]$ are the basis vectors in the crystal coordinate system. $\mathbf{e}_{1'}$, $\mathbf{e}_{2'}$ and $\mathbf{e}_{3'}$ are the basis vectors in the sample frame and $\mathbf{e}^{1'}$, $\mathbf{e}^{2'}$ and $\mathbf{e}^{3'}$ are the corresponding dual vectors.

For the tilt or twist GBs with the sample frame coinciding with the grain A frame, the coordinate transformation tensor $\mathbf{R} = R^{i'}_j \mathbf{e}_{i'} \otimes \mathbf{e}^j$ ($\mathbf{R} = \mathbf{g^{-1}}$) is determined with the rotation axis (ω_x, ω_y, ω_z) in crystal coordinate and angle ϕ (clockwise rotation) (Rodrigues' rotation formula) as

$$\left[R^{i'}_j\right] = \begin{bmatrix} \omega_x^2(1-\cos\phi) + \cos\phi & \omega_x\omega_y(1-\cos\phi) + \omega_z\sin\phi & \omega_x\omega_z(1-\cos\phi) - \omega_y\sin\phi \\ \omega_x\omega_y(1-\cos\phi) - \omega_z\sin\phi & \omega_y^2(1-\cos\phi) + \cos\phi & \omega_y\omega_z(1-\cos\phi) + \omega_x\sin\phi \\ \omega_x\omega_z(1-\cos\phi) + \omega_y\sin\phi & \omega_y\omega_z(1-\cos\phi) - \omega_x\sin\phi & \omega_z^2(1-\cos\phi) + \cos\phi \end{bmatrix} \tag{A.3}$$

with $\omega_x{}^2 + \omega_y{}^2 + \omega_z{}^2 = 1$. Then the slip vector $\mathbf{b}$ and normal vector $\mathbf{n}$ of the slip plane in the sample frame of grains A and B can be obtained as

$$\begin{cases} \mathbf{b}_{\mathrm{A}} = \mathbf{R} \cdot \mathbf{b}_{\mathrm{A}}^{\mathrm{c}}, & \mathbf{b}_{\mathrm{B}} = \mathbf{R} \cdot \mathbf{b}_{\mathrm{B}}^{\mathrm{c}}, \\ \mathbf{n}_{\mathrm{A}} = \mathbf{R} \cdot \mathbf{n}_{\mathrm{A}}^{\mathrm{c}}, & \mathbf{n}_{\mathrm{B}} = \mathbf{R} \cdot \mathbf{n}_{\mathrm{B}}^{\mathrm{c}}, \\ \mathbf{n}_{\mathrm{GB}} = \mathbf{R} \cdot \mathbf{n}_{\mathrm{GB}}^{\mathrm{c}} \end{cases} \tag{A.4}$$

with superscripts "c" representing vectors in the crystal frame.

Using Eq. (A.3), we could determine the transformation tensor $\mathbf{T} = T_j^{i'} \mathbf{e}_{i'} \otimes \mathbf{e}^j$ satisfying $\mathbf{T} \cdot \mathbf{b}_{\mathrm{A}} = \mathbf{b}_{\mathrm{B}}$, where the angle φ between $\mathbf{b}_{\mathrm{A}}$ and $\mathbf{b}_{\mathrm{B}}$ is given by Eq. (3.7) and the rotation axis $\mathbf{k} = \left(k_x, k_y, k_z\right)$ in crystal coordinate perpendicular to the plane composed by $\mathbf{b}_{\mathrm{A}}$ and $\mathbf{b}_{\mathrm{B}}$ is defined by Eq. (3.8). Under this definition, the tensor $\mathbf{T}$ can be expressed as

$$\left[T_j^{i'}\right] = \begin{bmatrix} k_x^2(1-\cos\varphi) + \cos\varphi & k_x k_y(1-\cos\varphi) - k_z \sin\varphi & k_x k_z(1-\cos\varphi) + k_y \sin\varphi \\ k_x k_y(1-\cos\varphi) + k_z \sin\varphi & k_y^2(1-\cos\varphi) + \cos\varphi & k_y k_z(1-\cos\varphi) - k_x \sin\varphi \\ k_x k_z(1-\cos\varphi) - k_y \sin\varphi & k_y k_z(1-\cos\varphi) + k_x \sin\varphi & k_z^2(1-\cos\varphi) + \cos\varphi \end{bmatrix}. \tag{A.5}$$

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